

My Method

$$[R_{200m}] = \frac{\text{kpc}}{h} \quad (\text{from SPARTA}) \quad (1)$$

$$[M_{200m}] = \frac{M_{\odot}}{h} \quad (\text{from } \text{colossus.mass_so.R.to.M}) \quad (2)$$

$$[G] = \frac{\frac{\text{kpc} \cdot \text{km}^2}{M_{\odot}}}{s^2} \quad (\text{from } \text{colossus.constants.G}) \quad (3)$$

$$[V_{200m}] = \frac{\text{km}}{s} \quad (\text{from } V_{200m} = \sqrt{\frac{G \cdot M_{200m}}{R_{200m}}}) \quad (4)$$

$$[t_{\text{dyn}}] = \frac{\text{kpc}}{[h] \cdot \text{km}} s \quad (\text{from } t_{\text{dyn}} = \frac{2 \cdot R_{200m}}{V_{200m}}) \quad (5)$$

$$[h] = \frac{\frac{\text{km}}{s}}{\text{Mpc}} \quad (\text{from SPARTA}) \quad (6)$$

(7)

To get final t_{dyn} value I multiplied by h as taken from SPARTA

Colossus Method

$$[t_{\text{hubb}}] = \frac{\text{Mpc}}{\frac{\text{km}}{s}} \quad (\text{from } 1/\text{cosmology.Hz}(z)) \quad (8)$$

$$[\rho_c] = \frac{M_{\odot} \cdot h^2}{\text{kpc}^3} \quad (\text{from } \text{cosmology.rho_c}(z)) \quad (9)$$

$$[\rho_m] = \frac{M_{\odot} \cdot h^2}{\text{kpc}^3} \quad (\text{from } \text{cosmology.rho_m}(z)) \quad (10)$$

$$[t_{\text{dyn}}] = \frac{\text{Mpc}}{\text{km}} s \quad (\text{from } t_{\text{dyn}} = 2^{\frac{3}{2}} t_{\text{hubb}} \left(\frac{\rho_m}{\rho_c} \right)^{-\frac{1}{2}}) \quad (11)$$

(12)

To get final t_{dyn} value I multiplied by 10^3 to get it in equivalent units of my method.

I found my method gave 2.600154416612781 while Colossus gave 5.306437645264576 which is pretty much exactly twice.